

Prefeasibility Study for Establishing an Amusement Park in Karak Governorate



2017



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1. Executive summary

The project aims to establish an amusement park in Karak governorate which is located in southern Jordan. Bounded from the South by Tafelah, and from the North by Madaba, it is 120 km away from Amman. Its mountains overlook the Dead Sea, the southern Jordan Valley and the West Bank in Palestine.

Karak is the sixth Jordanian governorate in terms of population in 2015, it is famous for its many archaeological sites such as Karak Castle, and its religious monuments such as maqamat ahsahaba, however; Karak lacks a amusement parks, so the idea of the project was to establish an amusement park that targets the governorate's residents, visitors and tourists inside and outside Jordan.

The project will increase tourism in the region and attract more tourists, in addition to providing 62 job opportunities for the governorate's residents.

Based on the market share and prices that have been adopted by the proposed project, the expected revenues will equal JD 1,363,500 during the first year of the project's lifetime, and will increase gradually to reach JD 2,322,239 in the tenth year. The payback period will be 5-years, and the internal rate of return will equal 27.19%.

Table1 : Economic indicators

Indicator	
Total Investment Requirement (JD)	1,101,578
Internal Rate of Return (IRR)	27.19%
Pay-Back Period (year)	5
Net present value (JD)	1,667,119
Job Opportunities provided by the	62
Annual Revenue Projections (10 years)	18,034,041
Total Operating Expenses (10 years)	13,196,272
Net Profit (10 years)	1,591,381

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

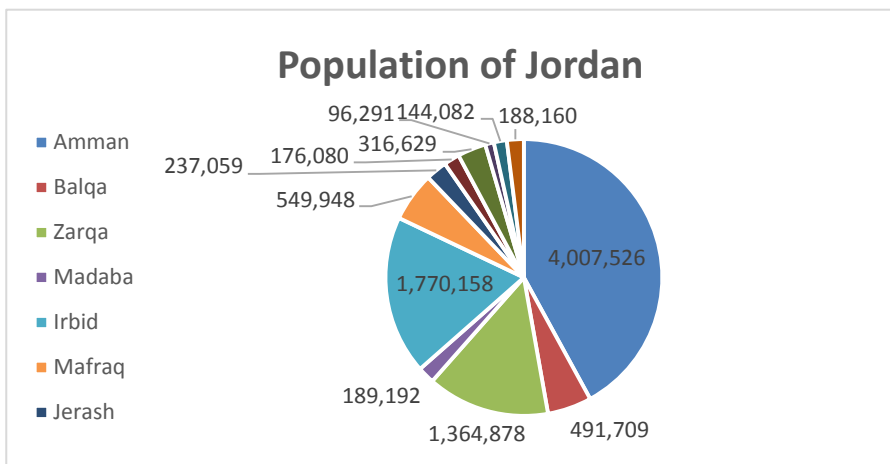
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1: Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30%

of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 : Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

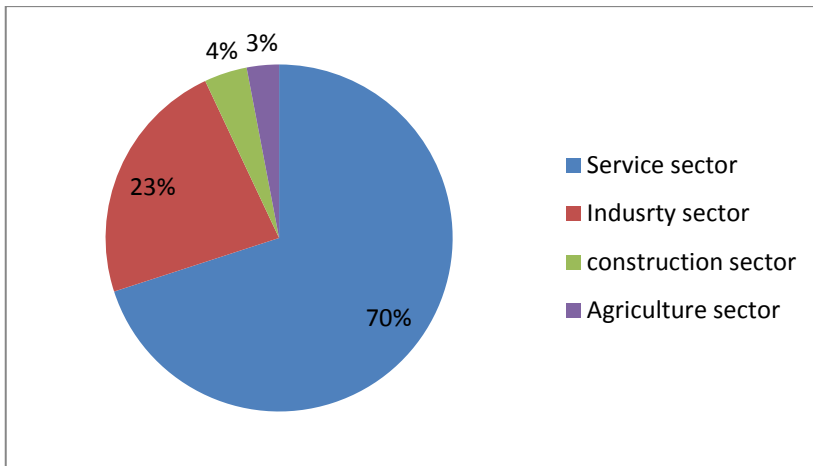
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2: Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is

dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce. Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- GDP Growth

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- Inflation rate

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- Unemployment

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- External Sector

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

Project description

The idea of the project is to establish an amusement park in Karak governorate which targets the governorate's residents and the rest of the Kingdom governorates and tourists. Since the governorate lacks any type of amusement parks like other tourist governorates the project's idea was adopted, where the project it will improve tourism in the region and attract more tourists.

The project consists of the following:

- A variety of games for children and adults including: (adults' games such as: The ship, "Hully Gully" the bride, "Tea Time" revolving saucer, "Mono rail without track" Russian mountain, the cupboard, the Dark Tower, chains, scissors, collision cars, the revolving vehicle, and the flying plate), children's games such as: (the worm, Rotary motor cars, chains for kids, the train, the Kangaroo, and collision cars for children).
- Ticket Booths.
- Sand areas for children
- Restaurant, coffee shop, fast food restaurants (cafeterias).
- Parking

Here are pictures of the games for clarification:

Table 3:games

The game	Picture	The game	Picture
The ship		"Tea Time" revolving saucer	
„Hully Gully” The bride		Russian mountain	
Dark tower		Chains	
The scissors		Collison cars	

The game	Picture	The game	Picture
Cupboard		The revolving vehical	
The revolving plate		The warm	
Rotary motor cars		Kids' chains	
The train		The kangaroo	

The game	Picture	The game	Picture
Collaisen cars for kids			

Project objectives

- The Project objectives are as follows:
- Achieve profit returns for the project owners
- Improve tourism in Karak by providing an amusement park project.
- Attract more visitors for the governorate
- Provide places for entertainment for the governorate's residents.
- Provide job opportunities

Proposed service

Provide an amusement park for Karak residents, visitors and to attract tourists to be the first park of its kind in the governorate.

Target segments

- The local community (mainly children and families)
- Karak visitors and tourists.

Market size analysis

The project's market is divided into two groups:

- The governorate local community (mainly children and families), as Karak population is 316,629 citizens.
- Tourists, on a second level, as the number of foreign tourists in Karak equals to 17,671 tourist, and 14,599 Jordanian visitors from the different kingdom's governorates in 2010, however; number of foreign tourists and visitors had decreased in 2015 to reach 12,633 tourist and 3900 visitors.

It is worth mentioning that the project depends mainly on the governorate's residents because of its lack of such projects. Therefore, the reduction in the number of tourists does not affect the project, furthermore, it is expected that the number of tourists will rise again after the end of the difficult conditions in the region, which were directly affecting the tourism sector.

Competitors analysis

This project is considered the first project of its kind in terms of size and amusement games, so there are no competitors for the project in the governorate, but there are some amusement parks that exist in other governorates, such as (Jibeha city) in Amman, and (Happy Land) in Irbid.

Capacity of the project

The project is an amusement park, therefore the total number of visitors is calculated by assuming that all games are full during the year and then calculating the percentage of the total capacity:

- Assumed occupancy rate for the amusement park: $416 \text{ (total seats in all games)} * 6 \text{ game/person/hour (assuming each game takes 10 minutes)} * 12 \text{ working hours/day} * 360 \text{ working days} = 10,800,000$
- The expected occupancy rate is equal to 2% of the total occupancy rate which is equal to 216,000 visitor/year.
- Percentage of expected game users from total visitors = 50% of total capacity = 108,000 games users for the first year.
- Facilities' users = 75% of the total expected number of visitors = 162,000 visitor for the first year

Price analysis and pricing policy

Based on the study of similar parks, the suggested prices are: Interance fees = JD 1/person and JD 0.5 per game/person.

Expected revenues

Annual expected revenues for the project:

- Income from interance fees: $1,080,000 \text{ total capacity} * 2\% \text{ target occupancy} * (\text{JD } 1) = \text{JD } 216,000$
- Income from games : $108,000 \text{ (gam users)} * (\text{JD } 0.5 / \text{ game}) * 216,000 \text{ (total number of visitors)} * 75\% \text{ (visitors percentage)} = \text{JD } 810,000$
- the following table shows the expected revenues for 10 years of the project start date.

Table 4: Project Revenues

Income resources	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Interance fees	1	1.03	1.06	1.09	1.13	1.16	1.19	1.23	1.27	1.30
Full capacity of the facility (person)	108000	108000	108000	108000	108000	108000	108000	108000	108000	108000
Target percentage from the total capacity	2.00%	2.044%	2.089%	2.135%	2.182%	2.230%	2.279%	2.329%	2.380%	2.433%
Total visitors (person)	216000	220752	225609	230572	235645	240829	246127	251542	257076	262731
Percentage of games users from total visitors	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%
Number of games users (person)	108000	110376	112804	115286	117822	120414	123063	125771	128538	131366
Game interance prices (JD)	0.5	0.52	0.53	0.55	0.56	0.58	0.60	0.61	0.63	0.65
Number of games/visitor	6	6.18	6.37	6.56	6.75	6.96	7.16	7.38	7.60	7.83
Facilities income JD / visitor	5	5.15	5.30	5.46	5.63	5.80	5.97	6.15	6.33	6.52
Percentage of facilities users	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%
Interance fees revenues (JD)	216000	227375	239348	251952	265220	279186	293888	309365	325656	342805
Games revenues (JD)	324000	351294	380887	412972	447761	485480	526377	570719	618796	670924
Facilities revenues (JD)	810000	852654.	897555	944821	994575	104694	110208	116011	122120	128551
Other revenues (JD)	13500	14313	15178	16097	17076	18116	19223	20402	21657	22992

4. Technical analysis

Required human resources

Table 5: Required human resources

Job title	Number
Indirect employees	
General manager	1
Accountant	1
Marketing officer	1
Security	10
Guard	2
Monitor and control officer	3
Maintenance	4
Total	22
Receptionist	4
Selling tickets employee	9
Games employees	18
Doctor	1
Nurse	1
Parking employees	2
Restaurant and cafeteria employees	5
Total	40
Grand total	62

The following shows a description for each job:

- General Manager: responsible for proposing general policies for the Amusement park, preparing private plans and following up their implementation, in addition to technical and administrative supervision of staff, developing procedures and ensuring occupational health and safety requirements.

- Accountant: following up financial affairs in addition to auditing.
- Marketing Officer: study market needs, pursue publicity activities, evaluating their effectiveness, and control sales and organization.
- Security Officer: maintaining security, protecting visitors within the park, and regulate the entry of visitors and any events in town.
 - Guard: Organizing visitor entry, in addition to close and open doors.
- Monitor and control: responsible for controlling electrical connections, and games technical situation and ensure games are running well.
- Maintenance officer: responsible for actions and activities of electric and mechanical maintenance, follow up the implementation of plans and programs established to ensure continuity of production process and quality at full capacity and lowest possible cost, and provide the necessary spare parts, and technical reports.
- Receptionist: receipt visitors, and make sure of collecting the required entry fees
- Tickets officer: visitors accounting at each game, and giving them tickets.
- Games officer: make sure to enter the right number of visitors to each game, play and pause the game.
- Doctor: handle any emergency injury occur in the park.
- Nurse: help the doctor and make patients ready for the doctor
- Parking Employee: park visitors' cars inside the parking.
- Cafeteria and restaurant employees: prepare snacks and sandwiches, serve customers, clean work tools, and apply occupational safety and health regulations.

Site analysis

For the preferred location of the project is a large piece of land near the population and serviced by all the necessary services. According to studies and assumptions, it is suggested that the project will be located in Zuhom area towards (Qatraneh) or (Southern Mizar in Mu'tah area), to be close to the University, where it is considered a tourist attraction as the historical site of mu'tah battle, and its proximity of the tafilah governorate.

Technical requirements analysis

1.1.1 Land and construction work

The project will be established on an 8,000 m² land, the following table shows the required land for the project:

Table6 : required land for the project

Land	Area (m ²)
Required land	8,000

The following table shows the required construction work for the project:

Table 7: Required construction

Construction work	Area (m ²)
Building work	100
Building design and schemes	400
Excavation work	100
Surface construction	100
Soil test	1
restaurant	200
cafeteria	30
Baths	100
Other	400

Machinery and equipment

The following table shows the required machinery and equipment's:

Table 8: Required machinery and equipment

Machinery and equipment	Quantity
The ship	1
The bride	1
Revolving saucer	1
Russian mountain	1
The Cupboard	1
The chains	1
Collision cars- adults	1
Revolving vehicle	1
Flying plate	1
The warm	1
Roller cars	1
Collision cars - children	1

Furniture and furnishings

The following table shows the require furniture and furnishings

Table 9: Required Furniture

Vehicles and transportation	Quantity
Pick-up	1

Furniture and furnishings	Quantity
Office furniture	3
TVs	5
Booths	5
Sweleih sand	2
Clinic supplies	1
Kitchens	1
Restaurant tables	30
Restaurant chairs	150
Restaurant cooking equipment	1

Vehicles and transportation

The following table shows the required Vehicles and transportation for the project:

Table 10: Required Vehicles and transportation



Vehicles and transportation	Quantity
Pick-up	1

5. Financial analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.23% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 11: General assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 12: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 13: Working Time

Working time	
Number of weeks/year	52
Number of working days/ week	7
Number of days of the year	360
Number of working hours/ day	8

Table 14: Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	18.50%
Utilities Expenses from Total Revenues	25.0%
Maintenance Expenses from Total Revenues	10%
Depreciation Expenses from Total Revenues	0.1%
Insurance Expenses from Total Assets Value	0.10%
Marketing Expenses from Total Revenues	2.5%

Table 16: Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	0%
Income Tax after Deduction	5%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	14

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	9.93%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 17: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%9.93
Income Tax Rate	%5
Bets	1.00
Equity	330,473
Loans Value	771,104
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	771,104
Loans Percentage	%70
Equity	
Cost of Equity	%16.4
Equity Value	330,473
Equity Percentage	%30
Gross	
Project Value	1,101,578
Weighted Average Cost of Capital	11.23%

Capital Expenditures

The estimated cost of the project is 1,101,578 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 18: Land Capital Expenditure

Land	Land	(Jordan Dinar/Meters Square) Cost	Area
Required Land	12.50	8,000	100,000
Cost			100,000

Table 19: Construction Work Capital Expenditure

Description	cost(JD/m2)	Area (m2)	Total
Construction Works	300	100	30,000
Design and blueprints	7	400	2,800
Excavation	5	100	500
Paving	12	100	1,200
Soil Test	200	1	200
Restaurant	300	200	60,000
Cafeteria	300	30	9,000
Bathrooms	300	100	30,000
Other Works	300.0	400	120,000
Total			253,700

Table 20: Machinery Capital expenses

Machinery and equipment	No.	Price/unit	Total
سفينة	1	25,000	25,000
عروسه	1	25,000	25,000
الصحن الدوار	1	6,000	6,000
الجبل الروسي	1	22,000	22,000
الدولاب	1	26,000	26,000
البرج المظلم	1	15,000	15,000
السلاسل	1	27,000	27,000
المقص	1	15,000	15,000
سيارات التصادم	1	8,000	8,000
لمركبه القلابه	1	22,000	22,000
الصحن الطائر	1	20,000	20,000
الدوده	1	20,000	20,000
السيارات الدواره	1	4,000	4,000
سلاسل الاطفال	1	7,000	7,000
القطار	1	10,000	10,000
الكنغر	1	7,000	7,000
سيارات تصادم اطفال	1	2,000	2,000
المجموع (دينار أردني)			261,000

Table 21: Furniture Capital expenses

	No.	Price/unit	Total (JD)
Office furniture	3	500	1,500
TVs	5	200	1,000
Booths	5	250	1,250
Sweleih sand	2	300	600
Clinic supplies	1	500	500
Kitchens	1	30,000	30,000
Restaurant tables	30	150	4,500
Restaurant chairs	150	50	7,500
Restaurant cooking equipment	1	1,000	1,000
other			47,850

Table 22: Transportation and Vehicles Capex

Transportation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Pick up	1	23,400	23,400
Total			23,400

Table 23: Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	1,500
Transportation	1,500
Marketing	7,500
Training	2,500
Total	13,000

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 24: Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	254,110
General and administrative expenses	3	%25	1,527
Marketing expenses	3	%25	8,522
Indirect Salaries	3	%25	38,325
Total			302,484

Table 25: Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Land	100,000	10%	110,000	%10.0
Construction works	253,700	10%	279,070	%25.3
Machinery & Equipment	261,000	10%	287,100	%26.1
Furniture	47,850	10%	52,635	%4.8
Vehicles	23,400	10%	25,740	%2.3
Pre-Operating Expenses	13,000	10%	14,300	%1.3
Working capital	302,484	10%	332,733	%30.2
Total (JD)	1,001,434		1,101,578	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,101,578) JD, where 70% of the project will be financed through loans (771,104) JD, and the remaining (30%) through (Equity) with (330,473) JD. The following table summarizes the general structure of the required financing:

Table 26: Sources of Funding

Funding Sources	Amount	Percentage
Equity	330,473	30%
Loans	771,104	70%

Total	1,101,578	100%
Use of Fund:	المبلغ	النسبة
Capital Expenditures	754,545	68.50%
Pre-Operating Expenses	14,300	1.30%
Working Capital	332,733	30.21%
Total	1,101,578	100%

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 27: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from	252,248	267,443	283,599	300,781	319,057	338,500	359,191	381,212	404,654	429,614
Utilities from Revenues	340,875	361,409	383,242	406,461	431,158	457,433	485,393	515,151	546,829	580,560
Maintenance from Revenues	136,350	144,564	153,297	162,584	172,463	182,973	194,157	206,060	218,732	232,224
Disposables from Revenues	1,364	1,446	1,533	1,626	1,725	1,830	1,942	2,061	2,187	2,322
Direct Salaries	193,200	201,121	209,367	217,951	226,887	236,190	245,873	255,954	266,448	277,373
Others	92,404	97,598	103,104	108,940	115,129	121,693	128,656	136,044	143,885	152,209
Total	1,016,44	1,073,58	1,134,14	1,198,34	1,266,41	1,338,61	1,415,21	1,496,48	1,582,73	1,674,30

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 28: General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	250	258	265	273	281	290	299	307	317	326
Transportation	750	773	796	820	844	869	896	922	950	979
Training	2,000	2,060	2,122	2,185	2,251	2,319	2,388	2,460	2,534	2,610
Insurance	755	777	800	825	849	875	901	928	956	985
Miscellaneous	555	572	589	607	625	644	663	683	704	725
Total	6,110	6,293	6,482	6,677	6,877	7,083	7,296	7,515	7,740	7,972

Marketing Expenses

Table 29: Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	34,088	36,141	38,324	40,646	43,116	45,743	48,539	51,515	54,683	58,056

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summarizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 30: Expected number of Employees

[illegible]

Total	40	40	40	40	40	40	40	40	40	40
Grand total	62	62	62	62	62	62	62	62	62	62

Table 31: Expected montly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
General manager	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Accountant	600	625	650	677	705	734	764	795	827	861
Marketing officer	900	937	975	1,015	1,057	1,100	1,145	1,192	1,241	1,292
Security	350	364	379	395	411	428	445	464	483	502
Guard	250	260	271	282	294	306	318	331	345	359
Monitor and control officer	450	468	488	508	528	550	573	596	621	646
Maintenance	400	416	433	451	470	489	509	530	552	574
Direct employees										
Receptionist	300	312	325	338	352	367	382	397	414	431
Selling tickets employee	350	364	379	395	411	428	445	464	483	502
Games employees	350	364	379	395	411	428	445	464	483	502
Doctor	700	729	759	790	822	856	891	927	965	1,005
Nurse	450	468	488	508	528	550	573	596	621	646
Parking employees	250	260	271	282	294	306	318	331	345	359

Restaurant and cafeteria employees	300	312	325	338	352	367	382	397	414	431
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Table 32: Expected Annual salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
General manager	35,000	36,435	37,929	39,484	41,103	42,788	44,542	46,369	48,270	50,249
Accountant	8,400	8,744	9,103	9,476	9,865	10,269	10,690	11,128	11,585	12,060
Marketing officer	12,600	13,117	13,654	14,214	14,797	15,404	16,035	16,693	17,377	18,090
Security	49,000	51,009	53,100	55,277	57,544	59,903	62,359	64,916	67,577	70,348
Guard	7,000	7,287	7,586	7,897	8,221	8,558	8,908	9,274	9,654	10,050
Monitor and control officer	18,900	19,675	20,482	21,321	22,195	23,106	24,053	25,039	26,066	27,134
Maintenance	22,400	23,318	24,274	25,270	26,306	27,384	28,507	29,676	30,893	32,159
Total	153,300	159,585	166,128	172,940	180,030	187,411	195,095	203,094	211,421	220,089
Direct employees										
Receptionist	16,800	17,489	18,206	18,952	19,729	20,538	21,380	22,257	23,169	24,119
Selling tickets employee	44,100	45,908	47,790	49,750	51,789	53,913	56,123	58,424	60,820	63,313
Games employees	88,200	91,816	95,581	99,499	103,579	107,826	112,247	116,849	121,639	126,627
Doctor	9,800	10,202	10,620	11,055	11,509	11,981	12,472	12,983	13,515	14,070
Nurse	6,300	6,558	6,827	7,107	7,398	7,702	8,018	8,346	8,689	9,045
Parking employees	7,000	7,287	7,586	7,897	8,221	8,558	8,908	9,274	9,654	10,050
Restaurant and cafeteria employees	21,000	21,861	22,757	23,690	24,662	25,673	26,725	27,821	28,962	30,149
Total	193,200	201,121	209,367	217,951	226,887	236,190	245,873	255,954	266,448	277,373

Grand total	346,500	360,707	375,495	390,891	406,917	423,601	440,969	459,048	477,869	497,462
Depreciations										

Table 33: Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	110,000	%0.0	%0.0
Construction Works	279,070	%5.0	%0.0
Machinaries	287,100	%10.0	%2.0
Furniture	52,635	%10.0	%5.0
Vehicles	25,740	%10.0	%0.0
Total	754,545		

Table 34: Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	754,545	762,919	771,539	780,414	789,554	798,969	808,667	818,660	828,959	839,575
Total Depreciation Expenses	50,501	51,338	52,200	53,088	54,002	54,943	55,913	56,913	57,942	59,004
Total Accumulated Depreciation	50,501	101,839	154,040	207,128	261,130	316,073	371,986	428,899	486,841	545,845
Total Additions	0	8,374	8,620	8,875	9,140	9,414	9,698	9,993	10,299	10,616

Total Net Book Values	704,044	661,079	617,499	573,287	528,425	482,896	436,681	389,762	342,118	293,730
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Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 35: Loan Details

Loan Value	771,104
Annual Interest Rate	9.48%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018
Annual Payment	200,720
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				771,104
2019	200,720	73,101	127,620	643,485
2020	200,720	61,002	139,718	503,767
2021	200,720	47,757	152,963	350,804
2022	200,720	33,256	167,464	183,340

2023	200,720	17,381	183,340	0
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Income Statement

Table 36: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	1,363,50	1,445,63	1,532,96	1,625,84	1,724,63	1,829,73	1,941,57	2,060,60	2,187,31	2,322,23
Gross Operating Revenues	1,363,50	1,445,63	1,532,96	1,625,84	1,724,63	1,829,73	1,941,57	2,060,60	2,187,31	2,322,23
Operating Expenses	1,016,44	1,073,58	1,134,14	1,198,34	1,266,41	1,338,61	1,415,21	1,496,48	1,582,73	1,674,30
Gross Operating Profit	347,060	372,056	398,826	427,500	458,213	491,114	526,360	564,122	604,582	647,937
<i>Gross Profit Percentage</i>	%25	%26	%26	%26	%27	%27	%27	%27	%28	%28
Salaries and Benefits (Indirect Staff)	-153,300	-159,585	-166,128	-172,940	-180,030	-187,411	-195,095	-203,094	-211,421	-220,089
General and Administraive Expenses	-6,110	-6,293	-6,482	-6,677	-6,877	-7,083	-7,296	-7,515	-7,740	-7,972
Markeing Expenses	-34,088	-36,141	-38,324	-40,646	-43,116	-45,743	-48,539	-51,515	-54,683	-58,056
Pre Operating Expenses	-14,300									
Gross Indirect Expenses	-207,797	-202,020	-210,935	-220,262	-230,023	-240,238	-250,930	-262,124	-273,844	-286,117
Income before Interest, Depreciation	139,263	170,036	187,892	207,237	228,190	250,876	275,430	301,998	330,738	361,819
Fixed Assets Depreciations	-50,501	-51,338	-52,200	-53,088	-54,002	-54,943	-55,913	-56,913	-57,942	-59,004

Income before Tax and Interests	88,762	118,698	135,691	154,149	174,188	195,932	219,517	245,086	272,796	302,815
Bank Interests	-73,101	-61,002	-47,757	-33,256	-17,381	-0	-0	-0	-0	-0
Income Before Tax	15,661	57,695	87,934	120,893	156,808	195,932	219,517	245,086	272,796	302,815
Income Tax	-783	-2,885	-4,397	-6,045	-7,840	-9,797	-10,976	-12,254	-13,640	-15,141
Net Profit	14,878	54,811	83,537	114,849	148,967	186,136	208,541	232,831	259,156	287,675
<i>Net Profit Percentage</i>	<i>%1</i>	<i>%4</i>	<i>%5</i>	<i>%7</i>	<i>%9</i>	<i>%10</i>	<i>%11</i>	<i>%11</i>	<i>%12</i>	<i>%12</i>
Compulsory Reserves	-1,488	-5,481	-8,354	-11,485	-14,897	-18,614	-20,854	-23,283	-25,916	-28,767
Retained Earnings	13,390	62,720	137,904	241,267	375,338	542,860	730,547	940,095	1,173,33	1,432,24

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 37: Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	14,878	54,811	83,537	114,849	148,967	186,136	208,541	232,831	259,156	287,675
Bank Interests	73,101	61,002	47,757	33,256	17,381	0	0	0	0	0
Depreciation	50,501	51,338	52,200	53,088	54,002	54,943	55,913	56,913	57,942	59,004
Total Operating Cashflows before	138,480	167,151	183,495	201,193	220,350	241,079	264,454	289,744	317,098	346,679
Inventory (Increase/Decrease)	-21,021	-1,266	-1,346	-1,432	-1,523	-1,620	-1,724	-1,835	-1,954	-2,080
Accounts Payables (Increase/Decrease)	127,365	6,991	7,401	7,837	8,300	8,793	9,317	9,875	10,469	11,101
Working Capital (Increase/Decrease)	106,345	5,725	6,054	6,405	6,777	7,172	7,593	8,040	8,515	9,021
Net Cashflows from Operating	244,825	172,876	189,549	207,597	227,127	248,252	272,047	297,784	325,614	355,700
Cashflows from Investments Activities										
Fixed Assets (Procurement)	-754,545	-8,374	-8,620	-8,875	-9,140	-9,414	-9,698	-9,993	-10,299	-10,616
Net Cashflows from Investments	-754,545	-8,374	-8,620	-8,875	-9,140	-9,414	-9,698	-9,993	-10,299	-10,616
Cashflows from Financing Activities										
Capital	330,473									
Loan Amortization	-127,620	-139,718	-152,963	-167,464	-183,340	0	0	0	0	0
Bank Interest Rate	-73,101	-61,002	-47,757	-33,256	-17,381	-0	-0	-0	-0	-0

Loans	771,104	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing	900,857	-200,720	-200,720	-200,720	-200,720	0	0	0	0	0
Net (Increase/Decrease) in Cash	391,137	-36,218	-19,791	-1,998	17,266	238,837	262,348	287,790	315,315	345,084
Cashflows at the Beginning of Period	0	391,137	354,919	335,128	333,130	350,396	589,233	851,582	1,139,372	1,454,687
Cashflows at the End of Period	391,137	354,919	335,128	333,130	350,396	589,233	851,582	1,139,372	1,454,687	1,799,771

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 38: Expected Balance Sheet

[illegible]

Non Current Assets	704,044	661,079	617,499	573,287	528,425	482,896	436,681	389,762	342,118	293,730
Fixed Assets (net)	704,044	661,079	617,499	573,287	528,425	482,896	436,681	389,762	342,118	293,730
Total Non Current Assets	1,116,202	1,038,285	976,260	931,481	905,409	1,100,337	1,318,195	1,560,901	1,830,526	2,129,302
Total Assets										
Liabilities	127,365	134,356	141,757	149,594	157,894	166,686	176,003	185,878	196,347	207,448
Current Liabilities	139,718	152,963	167,464	183,340	0-	0-	0-	0-	0-	0
Paybles	267,083	287,319	309,221	332,933	157,894	166,686	176,003	185,878	196,347	207,448
Remaining amount of Loan										
Total current Liabilities	503,767	350,804	183,340	0	0	0	0	0	0	0
Non Current Liabilities	503,767	350,804	183,340	0	0	0	0	0	0	0
Long Terms Loans	770,850	638,123	492,561	332,933	157,894	166,686	176,003	185,878	196,347	207,448
Total Long Term Liabilities										
Total Liabilities	330,473	330,473	330,473	330,473	330,473	330,473	330,473	330,473	330,473	330,473
Owners Equity	1,488	6,969	15,323	26,807	41,704	60,318	81,172	104,455	130,371	159,138
Shareholders Contributions	13,390	62,720	137,904	241,267	375,338	542,860	730,547	940,095	1,173,335	1,432,242
Statutory Reseve	345,351	400,162	483,700	598,548	747,515	933,651	1,142,192	1,375,023	1,634,179	1,921,854
Retained Profits	1,116,202	1,038,285	976,260	931,481	905,409	1,100,337	1,318,195	1,560,901	1,830,526	2,129,302

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 39: Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Total
Net Free Cashflows	-1,101,578	259,12	164,50	180,92	198,72	217,98	238,83	262,34	287,79	315,31	345,08	4,961,66
Discount Factor	1.00	0.81	0.73	0.65	0.59	0.53	0.47	0.43	0.38	0.34	0.31	0.31
Net Present Value for Cash	-1,101,578	209,43	119,52	118,18	116,70	115,08	113,36	111,94	110,39	108,74	106,99	1,538,32

Table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 40: Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-1,101,578	259,125	164,502	180,929	198,722	217,987	238,837	262,348	287,790	315,315	345,084
Project Value Returned	1,101,578	842,453	677,951	497,022	298,300	80,313	-158,524	-420,873	-708,663	-1,023,978	-1,369,062
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 41: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.23
Net Present Value for Cashflows	1,667,119
Payback Period	5
Internal Rate of Return	%27.19

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 42: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%27.19	5	11.23%
Revenues declined by 10%	%21.74	6	11.23%
Operating Expenses Increased by 10%	%15.50	9	11.23%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 1,667,119 Jordan Dinars considering that the project provides 62 Job Opportunities for the governorate residents.